Package 'waterlogged'

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Title Streamline the Processing of Water Logger Data

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Description This package aims to streamline the process of collating, converting and cleaning water chemistry data from continuous water loggers.

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Contents

add_nearest_baro 2
adjust_logger_NA 4
adjust_waterlevel_spike 5
adjust_WL_offset 7
adjust_WL_zero 8
barometric_qaqc_all 9
bind_hobo_files 11

conductivity_qaqc_all . . . 12
conductivity_temp_compensation . . . 14
convert_do_mgl_percsat . . . 15
convert_waterlevel_kPa_m . . . 16
get_logger_data . . . 18
NTSERP_palette . . . 19
plot_qaqc_timeseries . . . 20
QAQC_metadata . . . 21
QAQC_reference_data . . . 22
qc_airtemp . . . 23
qc_barometric . . . 24
qc_conductivity . . . 26
qc_dissox . . . 27
qc_waterlevel . . . 29
qc_watertemp . . . 30
resample_timestamps . . . 31
wl_multipoint_correction . . . 33
wl_set_datum . . . 34
wq_drift_correction . . . 36
write_clean_data . . . 38

`add_nearest_baro` *Attach nearest barometric pressure data to logger time series*

Description

Joins barometric pressure (`airpress_kPa`) and air temperature (`airtemp_C`) to water level, dissolved oxygen, or conductivity time series. The function operates in two modes:

Usage

```
add_nearest_baro(
  input_data,
  path_to_output_folder = NULL,
  baro_data_path = NULL,
  metric = c("water level", "DO", "conductivity", "both", "all"),
  max_km = NULL,
  max_stations = 5L,
  baro_site_selection = "auto",
  metadata_path = NULL,
  metadata = NULL,
  baro_data = NULL
)
```

Arguments

`input_data`	A data frame of logger observations. Must contain at minimum: `timestamp` (POSIXct or parseable by `lubridate::ymd_hms()`), `site_station_code`, and `metric`. Typically the output of `bind_hobo_files`.
`path_to_output_folder`	Character scalar. Root directory for v0.2 CSV output. If `NULL` (default), no files are written (Shiny mode).

`baro_data_path` Character scalar. Directory containing BARO CSV files (searched recursively). Required when `baro_data` is `NULL`.

`metric` Character. Which metrics require barometric compensation. One of `"water level"`, `"DO"`, `"conductivity"`, `"both"` (water level + DO), or `"all"`. Passed to `resolve_metrics_param` for normalisation.

`max_km` Numeric or `NULL`. Maximum search radius (km) for barometric stations in auto mode. Default is `NULL` (no distance limit — all barometric stations are considered). Set to a positive number to restrict candidates to stations within that radius.

`max_stations` Integer. Maximum number of nearest barometric stations to use per site in auto mode (default 5). Increasing this value reduces the chance of NA gaps when the closest stations have incomplete temporal coverage, at the cost of slightly more computation.

`baro_site_selection` Character. Either `"auto"` (default) for nearest-neighbour selection or a specific barometric `site_station_code` for manual selection.

`metadata_path` Character scalar. Path to a metadata CSV file. Required when `metadata` is `NULL`. The file is read and validated by `QAQC_metadata`.

`metadata` A data frame of station metadata (alternative to `metadata_path` for Shiny mode). Must contain at least: `site_station_code`, `latitude`, `longitude`, and `metric`. Exactly one of `metadata_path` or `metadata` must be supplied.

`baro_data` A data frame of barometric observations (alternative to `baro_data_path`). This is the recommended approach for cross-year workflows: run `barometric_qaqc_all` on your QAQC'd baro files and pass the combined output directly. Must contain: `site_station_code`, `timestamp`, and either `airpress_kPa` or `airpress_kPa_adj`. Optionally `airtemp_C` or `airtemp_C_adj`. Exactly one of `baro_data_path` or `baro_data` must be supplied.

Details

Auto mode (`baro_site_selection = "auto"`, the default.) Uses latitude/longitude from the metadata to find the nearest barometric stations (Haversine distance, up to `max_stations`). When `max_km` is non-`NULL`, candidates are limited to that radius. Overlapping series are calibrated to the nearest station via `calibrate_baro_series`, then merged into a single composite with `dplyr::coalesce()`.

Manual mode (`baro_site_selection = "<station_code>"`) Uses the specified barometric station directly, without distance filtering or calibration.

When `path_to_output_folder` is non-`NULL`, per-site, per-year v0.2 CSV files are written to `<path_to_output_folder>`. When `NULL` (Shiny mode), no files are written and the function is purely in-memory.

Auto-mode algorithm:

1. Compute Haversine distances between every site and every barometric station listed in the metadata.
2. Retain the nearest `max_stations` barometric stations (optionally filtered to within `max_km` when non-`NULL`).
3. For each site, read (or filter) barometric data for those stations, calibrate secondary/tertiary series to the nearest station using `calibrate_baro_series`, and `coalesce()` into a single composite series.
4. Left-join the composite onto the site data by `timestamp`.

The function prefers `_adj` columns (post-QAQC) when present: it selects `airpress_kPa_adj` over `airpress_kPa` and `airtemp_C_adj` over `airtemp_C`.

Timestamps are coerced to POSIXct (UTC) if not already.

Value

A two-element list:

`[[1]]` A data frame: the input rows augmented with columns `airpress_kPa`, `airtemp_C`, `baro_site_stn_code`, `baro_qaqc_adj`, `baro_qaqc_note`, and `baro_qaqc_code`.

`[[2]]` A `ggplot2` object: faceted time series of barometric pressure coloured by contributing baro station, for visual QC of the stitch.

See Also

`bind_hobo_files` (upstream, produces v0.1 data), `qc_barometric` and `barometric_qaqc_all` (QAQC baro data before this step), `resolve_metrics_param`, `calibrate_baro_series`

`adjust_logger_NA` *Set logger values to NA over a time window and log the change*

Description

Convenience wrapper to set adjusted logger values to `NA` over a specified time window (or, optionally, for all data below a dry threshold) for a given station and metric. Typical use cases are removing periods affected by ice, dry conditions, or logger disturbances after visual QA/QC.

Usage

```
adjust_logger_NA(
  input_data,
  select_station,
  metric,
  reason_to_adjust,
  timestamp_start,
  timestamp_end,
  keep_temp = FALSE,
  apply_to_all_data = FALSE,
  manual_note,
  log_root = NULL,
  user = NULL
)
```

Arguments

`input_data`	Data frame containing logger data for one or more stations, including a `timestamp` column, `site_station_code`, and the relevant adjusted fields (e.g. `waterlevel_m_adj`, `do_mgl_adj`, etc.).
`select_station`	Station code to adjust (value of `site_station_code`).

`metric` Character string specifying which metric to adjust. Accepted values include `"waterlevel"`, `"dissolvedoxygen"`, `"watertemp"`, `"barometric"`, `"airtemp"`, `"conductivity"` and common variants (e.g. `"water level"`, `"DO"`, `"BARO"`, `"AT"`).

`reason_to_adjust`
Character; reason for the adjustment. One of `"ice"`, `"dry"`, or `"disturbance"`. This controls which `edit_*` flag column is set and how the NA adjustment is applied. Note: barometric and air temperature metrics only accept `"disturbance"`.

`timestamp_start`, `timestamp_end`
Character date-time values defining the adjustment window, in `"YYYY-MM-DD HH:MM:SS"` format, parsed with `lubridate::ymd_hms()`.

`keep_temp` Logical; if `TRUE`, associated temperature adjustments are skipped for `"ice"` and `"disturbance"` events. Ignored for `"dry"` (temperature always removed) and for metrics where no temperature column is associated (`"watertemp"`, `"airtemp"`).

`apply_to_all_data`
Logical; if `FALSE` (default), only values within `timestamp_start`–`timestamp_end` are set to NA. If `TRUE` and `reason_to_adjust = "dry"`, a water level threshold is derived from the window and applied to the whole record. Requires a water level column (`waterlevel_m_adj` or `waterlevel_m`). Ignored for non-dry reasons and for metrics without water level support (`"watertemp"`, `"barometric"`, `"airtemp"`).

`manual_note` Character; required free-text explanation describing why this adjustment is being made. Stored in the QA/QC log.

`log_root` Root directory where QA/QC logs are stored (e.g. `"data/testing/processed"`). The actual log path is derived from this root, station, and metric.

`user` Character; user name recorded in the log entry. Defaults to `NULL`, which resolves to `Sys.info()[["user"]]` (the local OS user). In Shiny apps, pass the logged-in user explicitly (e.g. `session$user`).

Details

The function is metric-aware: it only modifies the adjusted fields relevant to the requested `metric` and sets a matching `edit_*` flag (e.g. `edit_remove_ice`, `edit_remove_dry`, `edit_remove_disturbance`). If no rows fall within the time window (or dry threshold) for the given station, a warning is issued and no log entry is written.

Value

A data frame for the selected station with updated adjusted fields and edit flags. Two internal helper columns (`in_window` and `edit_na`) are removed before returning.

See Also

`qc_waterlevel`, `qc_dissox`, `make_qaqc_log_row`, `qaqc_log_append`

`adjust_waterlevel_spike` *Flatten a water-level spike over a specified time window*

Description

Replaces a spike (or other unrealistic jump) in water level with a straight line between two timestamps for a single station, and records the change in the QA/QC log.

Usage

```
adjust_waterlevel_spike(
  input_data,
  select_station,
  timestamp_start,
  timestamp_end,
  reason_to_adjust,
  manual_note,
  log_root = NULL,
  user = NULL
)
```

Arguments

`input_data`	Data frame containing water-level data for one or more stations, including `timestamp`, `site_station_code`, `waterlevel_m`, and `waterlevel_m_adj`.
`select_station`	Character; station code to adjust (value of `site_station_code`).
`timestamp_start, timestamp_end`	Character date-time values defining the spike window, in `"YYYY-MM-DD HH:MM:SS"` format, parsed with `lubridate::ymd_hms()`.
`reason_to_adjust`	Character; reason for flattening. One of `"ice"` or `"disturbance"`. Used to choose the QA/QC code written to the log.
`manual_note`	Character; required free-text explanation describing why this adjustment is being made. Stored in the QA/QC log.
`log_root`	Root directory where QA/QC logs are stored (for example, `"data/testing/processed"`). The actual log path is derived from this root, station, and metric.
`user`	Character; user name recorded in the log entry. Defaults to `NULL`, which resolves to `Sys.info()[["user"]]` (the local OS user). In Shiny apps, pass the logged-in user explicitly (e.g. `session$user`).

Details

The function:

- Filters `input_data` to `select_station`.
- Locates the raw water level at `timestamp_start` and `timestamp_end`.
- Linearly interpolates between these two values over the full time window.
- Writes the interpolated values into `waterlevel_m_adj` for all timestamps in the window.

- Sets `edit_spike_flat = TRUE` for affected rows.
- Appends an entry to the QA/QC log via `make_qaqc_log_row()` and `qaqc_log_append()`.

The start and end water levels are taken from the raw `waterlevel_m` column at `timestamp_start` and `timestamp_end`. The interpolation is performed in time units of seconds and written into `waterlevel_m_adj`. If the function cannot find unique raw values at both timestamps, it errors.

Value

A data frame for `select_station` with updated `waterlevel_m_adj` values and `edit_spike_flat` flag. The input columns are preserved; only adjusted values and flags are changed.

See Also

`qc_waterlevel`, `adjust_logger_NA`, `make_qaqc_log_row`, `qaqc_log_append`

Examples

```
## Not run:
adjusted_wl <- adjust_waterlevel_spike(
  input_data      = waterlevel_complete_QAQC,
  select_station  = "WL_ALBR_ST_30",
  timestamp_start = "2025-02-11 21:00:00",
  timestamp_end   = "2025-02-20 16:00:00",
  reason_to_adjust = "ice",
  manual_note      = "Suspected ice formation due to freezing temps",
  log_root         = "data/testing/processed"
)

## End(Not run)
```

---

adjust_WL_offset *Apply a constant offset to water level after a break*

Description

Applies a constant offset to the adjusted water level series (`waterlevel_m_adj`) after a specified time, typically to correct for a cable break, logger shift, or other step change. The offset is computed from the difference between raw water levels at two reference timestamps and applied to all timestamps at or after `timestamp_end`. A QA/QC log entry is written for all affected rows.

Usage

```
adjust_WL_offset(
  input_data,
  select_station,
  timestamp_start,
  timestamp_end,
  manual_note,
  log_root = NULL,
  user = NULL
)
```

Arguments

input_data	Data frame containing water-level data for one or more stations, including timestamp, site_station_code, waterlevel_m, and waterlevel_m_adj.
select_station	Character; station code to adjust (value of site_station_code).
timestamp_start	Character date-time for the pre-offset reference point (in "YYYY-MM-DD HH:MM:SS" format), parsed with lubridate::ymd_hms().
timestamp_end	Character date-time for the post-offset reference point and the time from which the offset will be applied onward (same format as timestamp_start).
manual_note	Character; required free-text explanation describing why this adjustment is being made. Stored in the QA/QC log.
log_root	Root directory where QA/QC logs are stored (for example, "data/testing/processed"). The actual log path is derived from this root, station, and metric.
user	Character; user name recorded in the log entry. Defaults to NULL, which resolves to Sys.info()[["user"]] (the local OS user). In Shiny apps, pass the logged-in user explicitly (e.g. session$user).

Value

A data frame for select_station with updated waterlevel_m_adj values and edit_offset flag. The input columns are preserved; only adjusted values and flags are changed.

See Also

adjust_waterlevel_spike, adjust_logger_NA, make_qaqc_log_row, qaqc_log_append

Examples

```
## Not run:
station_wl_qc2 <- adjust_WL_offset(
  input_data      = station_wl_qc,
  select_station  = "WL_ALBR_ST_30",
  timestamp_start = "2025-02-04 07:00:00",
  timestamp_end   = "2025-02-09 00:00:00",
  manual_note     = "Animal suspected to have moved logger",
  log_root        = "data/testing/processed"
)

## End(Not run)
```

---

adjust_WL_zero *Adjust water level zero/reference*

Description

Shift adjusted water levels by a constant offset to change the zero/reference level. If change_value is NULL, the offset is computed from the mean water level in a specified time window (assumed to represent the new zero).

Usage

```
adjust_WL_zero(
  input_data,
  select_station,
  change_value = NULL,
  timestamp_start = NULL,
  timestamp_end = NULL,
  manual_note,
  log_root = NULL,
  user = NULL
)
```

Arguments

input_data	Data frame containing water level data for one or more stations. Must include timestamp, site_station_code, waterlevel_m, and waterlevel_m_adj.
select_station	Character; station code to adjust (value of site_station_code).
change_value	Numeric offset (m) to add to waterlevel_m_adj. Positive values shift levels up, negative values shift them down. If NULL, the offset is computed from the average water level between timestamp_start and timestamp_end.
timestamp_start	Start of window (character "YYYY-MM-DD HH:MM:SS", UTC). Required when change_value is NULL (used to compute the mean). When supplied with an explicit change_value, the offset is applied only within the window instead of the entire series.
timestamp_end	End of window (character "YYYY-MM-DD HH:MM:SS", UTC). See timestamp_start.
manual_note	Character, required. Description of why the zero shift was applied (stored in the QA/QC log).
log_root	Root folder used to build the QA/QC log path (passed to qaqc_log_path()).
user	Username written to the QA/QC log. Defaults to NULL, which resolves to Sys.info()[["user"]] (the local OS user). In Shiny apps, pass the logged-in user explicitly (e.g. session$user).

Details

When timestamp_start and timestamp_end are supplied, the offset is applied only within that window. Otherwise, the entire series is shifted.

If change_value is NULL, the function:

- Filters rows between timestamp_start and timestamp_end.
- Computes the mean of waterlevel_m_adj in that window.
- Uses the negative of this mean as the offset, so that the mean level in that window becomes approximately 0 m.

In all cases, edit_zero is set to TRUE for rows where the adjusted series was shifted, preserving any existing TRUE values from prior calls.

Value

Modified data frame for select_station with updated waterlevel_m_adj and edit_zero flag, and a QA/QC log entry written to disk.

See Also

adjust_WL_offset(), adjust_waterlevel_spike(), adjust_logger_NA(), make_qaqc_log_row(), qaqc_log_path(), qaqc_log_append()

---

barometric_qaqc_all *Run barometric QA/QC for all baro stations in metadata*

Description

Reads all barometric logger files for stations listed in the metadata (where metric == "barometric"), applies qc_barometric() to each station, writes yearly QA/QC CSVs (v0.1), and returns a combined data frame of all QA/QC'd barometric data.

Usage

```
barometric_qaqc_all(
  baro_data_path = NULL,
  metadata_path = NULL,
  path_to_output_folder = NULL,
  log_root = NULL,
  temp_low_limit = -40,
  temp_high_limit = 50,
  pressure_low_kpa = 85,
  pressure_high_kpa = 105,
  spike_threshold_kpa = 1.5,
  flatline_n = 6,
  user = NULL,
  metadata = NULL,
  baro_data = NULL
)
```

Arguments

baro_data_path	Root directory containing processed barometric CSV files. All subdirectories are searched recursively for files matching "<station>_BARO_*.csv", where <station> comes from the metadata. For example: "data/testing/processed".
metadata_path	Path to the metadata CSV used by QAQC_metadata(), e.g. "data/testing/raw/testing_metadata Must include site_station_code, metric, and barometric station definitions (metric == "barometric").
path_to_output_folder	Root directory where yearly QA/QC barometric files will be written. Year- and processed-subfolders are created as needed, e.g. "data/testing/processed".
log_root	Root folder where QA/QC logs are written by qc_barometric() and related functions, typically the same as path_to_output_folder, e.g. "data/testing/processed".
temp_low_limit	Lower bound (°C) for plausible barometric logger temperatures. Values outside this range may be flagged or removed.
temp_high_limit	Upper bound (°C) for plausible barometric logger temperatures.

`pressure_low_kpa`
: Lower bound (kPa) for plausible barometric pressure.

`pressure_high_kpa`
: Upper bound (kPa) for plausible barometric pressure.

`spike_threshold_kpa`
: Threshold (kPa) for flagging sudden pressure spikes between consecutive timesteps.

`flatline_n`
: Integer; number of consecutive identical values used to flag potential flatlining in the pressure series.

`user`
: Character username written into the QA/QC log. Defaults to `NULL`, which resolves to `Sys.info()[["user"]]` (the local OS user). In Shiny apps, pass the logged-in user explicitly (e.g. `session$user`).

`metadata`
: Optional data frame of station metadata (as returned by `QAQC_metadata`). When supplied, `metadata_path` is ignored and this data frame is used directly.

`baro_data`
: Optional data frame of pre-loaded barometric logger data. When supplied, `baro_data_path` is ignored and stations are filtered from this data frame instead of reading CSV files.

Details

Files are expected to follow the naming pattern `"<site_station_code>_BARO_YYYYMMDD_YYYYMMDD_v*.csv"` and live somewhere under `baro_data_path`.

Value

A single long-format data frame created by binding the QA/QC'd barometric data for all stations. If no stations are processed successfully, an empty tibble is returned and a warning is issued.

See Also

`qc_barometric()`, `add_nearest_baro()`

---

bind_hobo_files *Bind HOBO logger CSVs and attach site metadata*

Description

Reads all HOBO CSV files in a folder (or from explicit file paths), links each logger serial number to deployment metadata, trims records to deployment/removal windows, standardises core column names by metric (barometric, waterlevel, DO, TidbiT temperature, conductivity), optionally writes per-station/year v0.1 CSVs, and returns a compiled long data frame plus a quick QC plot.

Usage

```
bind_hobo_files(
  path_to_raw_folder = NULL,
  path_to_output_folder = NULL,
  metadata_path = NULL,
  timestamp_timezone = "UTC",
  file_paths = NULL,
  metadata = NULL
)
```

Arguments

`path_to_raw_folder`
: Character scalar. Path to the folder containing raw HOBO CSV files (one or more loggers, all of the same metric). For example `"data/raw/level"` or `"data/testing/raw/COND"`. Required when `file_paths` is `NULL`.

`path_to_output_folder`
: Character scalar or `NULL`. Root folder where processed v0.1 CSVs will be written. Files are created under `<path_to_output_folder>/<year>/processed/`. When `NULL` (default, Shiny mode), no files are written.

`metadata_path`
: Character scalar. Path to a metadata CSV that has at least logger serial numbers and deployment/removal information (e.g. `sn`, `metric`, `site_station_code`, `timestamp_deploy`, `timestamp_remove`, `model`). Required when `metadata` is `NULL`.

`timestamp_timezone`
: Character scalar. Time zone used when filling in open-ended deployments (where `timestamp_remove` is `NA`). Defaults to `"UTC"`.

`file_paths`
: Character vector of full file paths to HOBO CSV files (alternative to `path_to_raw_folder` for Shiny mode). Exactly one of `path_to_raw_folder` or `file_paths` must be supplied.

`metadata`
: A data frame of station metadata (alternative to `metadata_path` for Shiny mode). Must contain the same columns expected by `QAQC_metadata`. Exactly one of `metadata_path` or `metadata` must be supplied.

Details

The function:

- Reads all `*.csv` files in `path_to_raw_folder` (or uses `file_paths` directly) via `extract_alldata_from_file()` and binds them into a single data frame.
- Uses `QAQC_metadata` to clean/standardise the metadata.
- Assigns `site_station_code` to each record based on `sn` and the deployment/removal timestamps.
- Enforces a single `metric` per folder (e.g., all barometric, all waterlevel, all conductivity).
- Standardises core column names depending on logger/metric, e.g. `airpress_kPa`, `waterpress_kPa`, `watertemp_C`, `airtemp_C`, `do_mgl`, `conduct_uScm`, and `conduct_source`.
- For conductivity (U24) loggers, unifies low/full range columns into a single `conduct_uScm` column and records which range was used in `conduct_source`.
- When `path_to_output_folder` is non-`NULL`, writes one v0.1 CSV per station per year to `<path_to_output_folder>/<year>/processed/` with filenames of the form `<site>_<LOGGER>_<startdate>_<`

Value

A list of length 2:

`[[1]]` A data frame of compiled logger data (`sites_compiled`), with `site_station_code`, `sn`, `timestamp`, `logger_type`, `metric`, and metric-specific value columns.

`[[2]]` A `ggplot2` object showing the primary value column vs time, faceted by `site_station_code`, for quick visual QA/QC.

See Also

add_nearest_baro (downstream, attaches baro data to produce v0.2), barometric_qaqc_all (QAQC baro data produced by this function), QAQC_metadata (metadata validation)

`conductivity_qaqc_all` *Run conductivity QAQC for all conductivity stations in metadata*

Description

Reads conductivity logger data for every station listed in the metadata (where `metric == "conductivity"`), applies qc_conductivity to each station, optionally writes per-year v0.1 CSVs, and returns a combined data frame of all QAQC'd conductivity data.

Usage

```
conductivity_qaqc_all(
  cond_data_path = NULL,
  metadata_path = NULL,
  path_to_output_folder = NULL,
  log_root = NULL,
  temp_low_limit_C = -2,
  temp_high_limit_C = 40,
  disturbance_threshold_uScm = 200,
  dry_threshold_uScm = 10,
  air_water_diff_threshold_C = 2,
  file_suffix = "COND",
  user = NULL,
  metadata = NULL,
  cond_data = NULL
)
```

Arguments

`cond_data_path`	Character scalar or `NULL`. Root directory containing processed conductivity CSV files (searched recursively). Required when `cond_data` is `NULL`.
`metadata_path`	Character scalar or `NULL`. Path to the metadata CSV. Required when `metadata` is `NULL`. Must include `site_station_code` and `metric` columns with entries where `metric == "conductivity"`.
`path_to_output_folder`	Character scalar or `NULL`. Root directory where per-year v0.1 COND CSVs will be written. When `NULL` (default, Shiny mode), no files are written.
`log_root`	Character scalar or `NULL`. Root directory for QAQC log files, passed through to qc_conductivity. When `NULL` (default), no log files are written.
`temp_low_limit_C`	Numeric. Lower bound for acceptable water temperature (passed to qc_conductivity). Default `-2`.
`temp_high_limit_C`	Numeric. Upper bound for acceptable water temperature. Default `40`.
`disturbance_threshold_uScm`	Numeric. Spike detection threshold between consecutive samples. Default `200`.

dry_threshold_uScm
: Numeric. Conductivity below this value flags a probable dry sensor. Default 10.

air_water_diff_threshold_C
: Numeric. Maximum air–water temperature difference to support dry inference. Default 2.

file_suffix
: Character scalar. Suffix used when building the filename search pattern (e.g. "COND" matches STATION_COND_*.csv). Default "COND".

user
: Character scalar. User name recorded in QAQC logs. Defaults to NULL, which resolves to Sys.info()[["user"]] (the local OS user). In Shiny apps, pass the logged-in user explicitly (e.g. session$user).

metadata
: A data frame of station metadata (alternative to metadata_path for Shiny mode). Exactly one of metadata_path or metadata must be supplied.

cond_data
: A data frame of conductivity observations (alternative to cond_data_path for Shiny mode). Must contain site_station_code, timestamp, conduct_uScm, and watertemp_C. Exactly one of cond_data_path or cond_data must be supplied.

Details

When reading from disk, files are expected to match the pattern "<site_station_code>_<file_suffix>_*.csv" under cond_data_path (searched recursively).

Value

A single long-format data frame created by binding the QAQC'd conductivity data for all stations. If no stations are processed successfully, an empty tibble is returned with a warning.

See Also

qc_conductivity (per-station QAQC called internally), barometric_qaqc_all (analogous batch wrapper for barometric data), bind_hobo_files (upstream data ingestion)

---

conductivity_temp_compensation

Temperature-compensate conductivity to specific conductance

Description

Converts measured electrical conductivity (EC) to specific conductance (SPC) normalised to 25 degrees C using the temperature compensation factors from ISO 27888. A 5th-order polynomial is fitted to the published coefficient table and applied as SPC = f(T) * EC.

Usage

```
conductivity_temp_compensation(input_data)
```

Arguments

input_data
: A data frame of conductivity observations. Must contain at least one of conduct_uScm_adj or conduct_uScm (electrical conductivity in uS/cm) and at least one of watertemp_C_adj or watertemp_C (water temperature in degrees C).

Details

Can be run either before or after qc_conductivity. The function prefers *_adj columns when available and falls back to the raw columns (conduct_uScm, watertemp_C) otherwise.

The compensation coefficients are from ISO 27888, tabulating the correction factor at 0.1 degrees C resolution from 0 to 35 degrees C. The coefficients are embedded directly in the function source. A 5th-order polynomial $f(T)$ is fitted by ordinary least squares. At 25 degrees C the coefficient equals 1.0 (i.e. no correction).

Value

input_data with one column added:

spc_uScm_adj Specific conductance (uS/cm at 25 degrees C). NA where either the conductivity or temperature input is NA.

A warning is issued when more than 5\ NA.

See Also

qc_conductivity, wq_drift_correction (optional drift correction for conductivity)

---

convert_do_mgl_percsat

Convert dissolved oxygen (mg/L) to percent air saturation

Description

Computes percent air saturation from dissolved oxygen concentrations (do_mgl), water temperature (watertemp_C), and barometric pressure (airpress_kPa), following standard Weiss (1970/1974) oxygen solubility equations.

Usage

```
convert_do_mgl_percsat(
  input_data,
  path_to_output_folder = NULL,
  version_label = "v0.3"
)
```

Arguments

input_data	A data frame containing DO, temperature, barometric pressure, timestamp, and site identifiers. Must contain the columns: timestamp, site_station_code, do_mgl, watertemp_C, and airpress_kPa.
path_to_output_folder	Character scalar. Root folder where per-year, per-site output files will be written. Must already exist.
version_label	Character version string appended to output filenames (e.g., "v0.3").

Details

The function also writes per-site, per-year CSV files using the naming:

```
<site>_DO_<startdate>_<enddate>_<version>.csv
```

and stores them in:

```
<path_to_output_folder>/<year>/processed/
```

Input data must already be QA/QC'd and standardized to metric-based column names:

- `timestamp` (POSIXct, UTC)
- `site_station_code`
- `do_mgl` dissolved oxygen (mg/L)
- `watertemp_C` water temperature (deg C)
- `airpress_kPa` barometric pressure (kPa)

Values of `-888.88` in `do_mgl` are treated as missing.

The calculation uses Weiss (1970, 1974) oxygen solubility equations. Barometric pressure is internally converted from kPa to atmospheres.

Value

A data frame identical to `input_data` but with an additional numeric column `do_percsat` (percent air saturation). Files are written to disk as side effects.

References

Weiss, R. F. (1970). *The solubility of nitrogen, oxygen and argon in water and seawater.* Deep-Sea Research.

Weiss, R. F. (1974). *Oxygen solubility in water and seawater based on the virial equation.* Deep-Sea Research.

`convert_waterlevel_kPa_m` *Convert water level from kPa to metres via barometric compensation*

Description

Converts raw water-pressure logger readings (kPa) to water level in metres using barometric compensation and a manual stage or depth reference measurement. This is the v0.3 step in the processing pipeline, typically run after `add_nearest_baro` (v0.2) has attached barometric data.

Usage

```
convert_waterlevel_kPa_m(
  input_data,
  select_station = "all",
  reference_data,
  reference_type = "stage",
  select_measurement = 1,
  logger_type_expected = "u20",
  path_to_output_folder = NULL,
  return_plots = TRUE
)
```

Arguments

input_data	A data frame of logger observations. Must contain: site_station_code, timestamp (POSIXct or parseable by lubridate::ymd_hms()), waterpress_kPa, watertemp_C, and airpress_kPa. Optionally logger_type (used for a sanity check against logger_type_expected). Typically the output of add_nearest_baro.
select_station	Character scalar, "all", or NA. When a single station code is supplied, only that station is processed. When "all", NA, NULL, or missing, every unique site_station_code in input_data is processed in batch mode; failures for individual stations produce warnings rather than errors.
reference_data	Character scalar (file path) or data frame. Manual stage/depth reference measurements. When a character path is supplied, the file is read and validated by QAQC_reference_data. When a data frame is supplied (Shiny mode), it is validated directly. Must contain site_station_code, timestamp, and at least one of stage_m or depth_m (matching reference_type).
reference_type	Character scalar: "stage" (default) or "depth". Determines which reference column (stage_m or depth_m) is used for the barometric compensation constant.
select_measurement	Integer scalar. Index (1-based) of the reference measurement to use as the anchor point for computing the compensation constant k. Default 1 (first in-range measurement).
logger_type_expected	Character scalar. Expected pattern in the logger_type column (matched case-insensitively). Default "u20" (HOBO U20 water-level logger).
path_to_output_folder	Character scalar or NULL. Root directory where per-year v0.3 WL CSVs will be written. When NULL (default, Shiny mode), no files are written.
return_plots	Logical; if TRUE (default), an interactive **plotly** chart is included in the returned list. Set to FALSE to skip plot creation (faster, useful in Shiny or batch pipelines).

Details

The conversion proceeds as follows:

1. **Reference data validation.** QAQC_reference_data validates the reference measurements (timestamp parsing, numeric coercion, duplicate detection).
2. **Input data validation.** QAQC_wl_kpa_inputs() checks required columns, parses timestamps, and verifies logger_type.

3. **Fluid density.** Temperature-dependent water density is calculated using the IAPWS-recommended polynomial (Kell, 1975) and converted to lb/ft^3 for intermediate unit conversion.
4. **Pressure to depth.** Water and barometric pressures (kPa) are converted to equivalent depth (m) via the relation `D = 0.3048 * (P * 0.1450377 * 144) / rho`, where `rho` is in lb/ft^3.
5. **Compensation constant *k*.** At the reference measurement selected by `select_measurement`, the constant $k = L - (D_{sensor} - D_{baro})$ is computed, where L is the manual stage or depth reading.
6. **Water level.** The full time series is computed as $WL = D_{sensor} - D_{baro} + k$.

When `path_to_output_folder` is non-`NULL`, per-year v0.3 CSVs are written to `<path_to_output_folder>/<year>/p`

Value

A three-element named list:

`site_wl` Data frame of logger observations with added columns: `density_kgm3`, `density_lbft3`, `sensor_depth_m`, `baro_sensor_depth_m`, `waterlevel_m`, `waterlevel_reference`, and `ref_m` (joined reference values, `NA` where no reference exists). In batch mode the data frames for all stations are row-bound.

`ref_dat` Data frame of reference measurements used (`timestamp`, `ref_m`, `site_comments`). In batch mode the reference data for all stations are row-bound.

`wl_plot` An interactive **plotly** object showing the converted water-level time series (dark line) overlaid with reference measurements (orange points). In batch mode the plot is facetted by station.

See Also

`add_nearest_baro` (upstream, attaches barometric data), `QAQC_reference_data` (validates reference measurements), `wl_set_datum` (downstream, sets arbitrary datum), `qc_waterlevel` (downstream QAQC)

`get_logger_data` *Retrieve and summarise logger data*

Description

High-level helper to load logger data from disk, optionally filter by station, optionally summarise to a temporal scale, and optionally return plots.

Usage

```
get_logger_data(
  path_to_data = NULL,
  data_processing = "v1.0",
  metric = NULL,
  select_station = "all",
  temporal_scale = "none",
  return_plots = TRUE,
  input_data = NULL
)
```

Arguments

path_to_data	Path to the folder containing the CSV files. The function will search this folder and its subfolders recursively.
data_processing	One of "raw", "v0.1", "v0.2", "v0.3", "v1.0", "clean". The value is used to select files by version suffix. "clean" is treated as "v1.0". Summarisation (i.e., temporal_scale != "none") is only allowed for v1.0/clean data.
metric	A character string indicating which metric group to summarise. Used only when temporal_scale != "none". Accepted values (case and spaces ignored) are mapped to: • "waterlevel": water level + water temperature • "dissolvedoxygen": DO (mg/L, %sat) + water temperature • "barometric": barometric pressure + air temperature • "conductivity": conductivity + water temperature • "airtemp": air temperature (e.g., TidbiT) • "watertemp": water temperature (e.g., TidbiT)
select_station	Either "all" (default) or one or more site_station_code values to filter on.
temporal_scale	Temporal aggregation. Case and spaces ignored and normalised to one of: "none", "hourly", "daily", "weekly", "monthly", "yearly". Use "none" to return raw timestamps with no summarisation. "Hourly" aggregates to average-by-hour (using floor_date).
return_plots	Logical; if TRUE and temporal_scale != "none", returns a list with data and ggplot objects. If FALSE, returns just the data frame.
input_data	A data frame of logger observations (alternative to path_to_data for Shiny mode). When supplied, file discovery is skipped and this data frame is used directly. Exactly one of path_to_data or input_data must be provided.

Details

This function works with the original wide-format processed files (e.g. columns like waterlevel_m_U20_adj, do_mgl_U26_adj). Summarisation is only performed for clean data (v1.0), and only for the measurement variables corresponding to the chosen metric.

Value

- If temporal_scale == "none": a data frame of combined logger data.
- Otherwise, if return_plots = FALSE: a summarised data frame.
- Otherwise, a list with elements:
  - data: summarised data frame
  - plot1, plot2, plot3: ggplot objects (some may be NULL depending on metric).

See Also

write_clean_data (produces the v1.0 files this function reads), bind_hobo_files (produces v0.1), plot_qaqc_timeseries (interactive QC plotting)

NTSERP_palette *Color-blind Friendly Palette for NTSERP lab*

Description

A function that asks for the number of colors (groups) needed and then gives a visually tested color palette for use in figures.

Usage

```
NTSERP_palette(n_groups = 0, full_palette = FALSE)
```

Arguments

n_groups	the number of colors (groups) required
full_palette	logical; if TRUE, the full palette is returned (not recommended for plots)

Value

list of colors suitable for colorblind viewing

plot_qaqc_timeseries *Plot QA/QC time series for logger data*

Description

Creates a multi-panel plot showing water level (WL), dissolved oxygen (DO), barometric pressure (BARO), conductivity (COND), standalone air temperature (AT), and/or standalone water temperature (WT) time series (raw vs. adjusted), corresponding logger/air temperatures, and QA/QC flags.

Usage

```
plot_qaqc_timeseries(
  wl_data = NULL,
  do_data = NULL,
  baro_data = NULL,
  cond_data = NULL,
  airtemp_data = NULL,
  watertemp_data = NULL,
  select_station,
  static = FALSE
)
```

Arguments

wl_data	Optional data frame of water-level records.
do_data	Optional data frame of dissolved-oxygen records.
baro_data	Optional data frame of barometric records. If NULL and WL data contain barometric columns, those are used.
cond_data	Optional data frame of conductivity records.
airtemp_data	Optional data frame of standalone air temperature records (e.g. output of qc_airtemp). Expected columns: airtemp_C, airtemp_C_adj, and optionally edit_temp_range.
watertemp_data	Optional data frame of standalone water temperature records (e.g. output of qc_watertemp). Expected columns: watertemp_C, watertemp_C_adj, and optionally flag/edit columns flag_ice, flag_wt_dry, edit_temp_lt_minus1, edit_temp_neg_near.
select_station	Character station code to plot.
static	Logical. If FALSE (default), returns an interactive Plotly htmlwidget with a range-slider for navigation. If TRUE, returns a static patchwork / gg object suitable for PDF output. Requires the **patchwork** package.

Details

Each data set is filtered to select_station, and any available QA/QC flags are shown in separate panels.

Value

When static = FALSE, a Plotly htmlwidget. When static = TRUE, a patchwork object (inherits gg).

Examples

```
## Not run:
plot_qaqc_timeseries(
  wl_data   = waterlevel_complete_QAQC,
  do_data   = station_do_dat,
  baro_data = all_checked_baro_data,
  cond_data = station_cond_dat,
  select_station = "ALBR_ST_30"
)

# standalone temperature plots
plot_qaqc_timeseries(
  airtemp_data   = qc_airtemp_result,
  watertemp_data = qc_watertemp_result,
  select_station = "ALBR_ST_30"
)

## End(Not run)
```

`QAQC_metadata` *QA/QC for logger deployment metadata*

Description

Reads (from file or in-memory data frame) and validates a table of logger deployment metadata used throughout the **waterlogged** pipeline. Called internally by `bind_hobo_files`, `add_nearest_baro`, `barometric_qaqc_all`, and `conductivity_qaqc_all`.

Usage

```
QAQC_metadata(metadata_path = NULL, metadata = NULL, tz = "UTC")
```

Arguments

`metadata_path`	Character scalar. Path to a metadata CSV file. Required when `metadata` is `NULL`.
`metadata`	A data frame of deployment metadata (alternative to `metadata_path` for Shiny mode). Exactly one of `metadata_path` or `metadata` must be supplied.

Details

The following validation steps are applied in order:

1. Empty rows (all `NA` or blank) are dropped.
2. Required columns are checked: `site_station_code`, `timestamp_deploy`, `timestamp_remove`, `model`, `sn`, `metric`, `status`, `latitude`, `longitude`.
3. `timestamp_deploy` and `timestamp_remove` are parsed using `lubridate::mdy_hm()` first, then fall back to flexible multi-format parsing. `NA` values in `timestamp_remove` are expected for loggers still deployed.
4. Chronology is checked: deployment must precede removal.
5. `sn` (serial number) is coerced to character.
6. Latitude and longitude are validated: DMS / cardinal-letter formats trigger a warning, values are coerced to numeric, and range checks ($[-90, 90]$ and $[-180, 180]$) are applied.
7. `metric` is standardised to lowercase with spaces removed and validated against the allowed set: `"waterlevel"`, `"barometric"`, `"conductivity"`, `"dissolvedoxygen"`, `"airtemp"`, `"watertemp"`, `"light"`.

Value

A cleaned data frame with:

- `timestamp_deploy` and `timestamp_remove` parsed to POSIXct (UTC).
- `latitude` and `longitude` coerced to numeric decimal degrees.
- `sn` coerced to character (preserves leading zeros).
- `metric` standardised to lowercase with spaces removed.

See Also

`QAQC_reference_data` (analogous validation for manual stage/depth reference data), `bind_hobo_files` (primary consumer of the validated metadata)

QAQC_reference_data *QA/QC for manual stage/depth reference data*

Description

Reads (from file or in-memory data frame) and validates a table of manual water-level measurements (stage or depth) used as reference levels for the kPa-to-metres barometric compensation performed by convert_waterlevel_kPa_m.

Usage

```
QAQC_reference_data(ref_path = NULL, ref_data = NULL)
```

Arguments

ref_path	Character scalar. Path to a reference CSV file. Required when ref_data is NULL.
ref_data	A data frame of reference measurements (alternative to ref_path for Shiny mode). Exactly one of ref_path or ref_data must be supplied.

Details

The following validation steps are applied in order:

1. Empty rows (all NA or blank) are dropped.
2. Required columns are checked: site_station_code, timestamp, site_comments, and at least one of stage_m or depth_m.
3. timestamp is parsed using lubridate::mdy_hm() first, then falls back to flexible multi-format parsing (mdy HM, ymd HMS, etc.).
4. stage_m and depth_m are coerced to numeric; negative values trigger a warning.
5. If no usable stage, depth, or discharge values exist, a warning is issued.
6. Duplicate (site_station_code, timestamp) pairs are flagged with a warning.

Value

A cleaned data frame with:

- timestamp parsed to POSIXct (UTC).
- stage_m and/or depth_m coerced to numeric.
- site_station_code and site_comments trimmed of whitespace.

See Also

convert_waterlevel_kPa_m (consumes the validated reference data), QAQC_metadata (analogous validation for deployment metadata)

qc_airtemp *QA/QC a single air temperature time series*

Description

Applies automatic range-check QA/QC to an air temperature logger time series for one station. Out-of-range values are set to NA in the adjusted column and flagged. Returns the input data (filtered to select_station) with adjusted columns and a boolean edit flag.

Usage

```
qc_airtemp(
  input_data,
  select_station,
  temp_low_limit = -40,
  temp_high_limit = 40,
  log_root = NULL,
  user = NULL
)
```

Arguments

input_data	Data frame of barometric or air-temperature logger data containing at least timestamp, site_station_code, and airtemp_C.
select_station	Character, station code to process (matches site_station_code).
temp_low_limit	Lower bound (C) for plausible air temperature. Values below this are set to NA in airtemp_C_adj and flagged.
temp_high_limit	Upper bound (C) for plausible air temperature. Values above this are set to NA in airtemp_C_adj and flagged.
log_root	Character scalar or NULL. Root directory for QAQC log files. When non-NULL, a log CSV is written via qaqc_log_maybe_append(). When NULL (default, Shiny mode), no file is written.
user	Character username written into the QA/QC log. Defaults to NULL, which resolves to Sys.info()[["user"]].

Details

If airtemp_C_adj does not exist it is created as a copy of airtemp_C. OUTLIER_AT is created if missing and set to TRUE where the rule applies.

Value

A data frame containing only rows for select_station, sorted by timestamp, with airtemp_C_adj and OUTLIER_AT columns added. attr(result, "qaqc_log") holds a tibble of QA/QC log rows.

qc_barometric *QA/QC a single barometric pressure time series*

Description

Applies automatic QA/QC rules to a barometric logger time series for one station: removes out-of-range pressures and temperatures, flags spikes and flatlines, and writes QA/QC log entries. Returns the input data with adjusted columns (`airpress_kPa_adj`, `airtemp_C_adj`) and boolean edit flags.

Usage

```
qc_barometric(
  input_data,
  select_station,
  log_root = NULL,
  user = NULL,
  temp_low_limit = -40,
  temp_high_limit = 50,
  pressure_low_kpa = 51,
  pressure_high_kpa = 112,
  spike_threshold_kpa = 1.5,
  flatline_n = 6
)
```

Arguments

`input_data`	Data frame of barometric logger data containing at least `timestamp`, `site_station_code`, `airpress_kPa` and `airtemp_C`, possibly for multiple stations.
`select_station`	Character, station code to process (matches `site_station_code`).
`log_root`	Character scalar or `NULL`. Root directory for QAQC log files. When non-`NULL`, a log CSV is written via `qaqc_log_maybe_append()`. When `NULL` (default, Shiny mode), no file is written.
`user`	Character username written into the QA/QC log. Defaults to `NULL`, which resolves to `Sys.info()[["user"]]` (the local OS user). In Shiny apps, pass the logged-in user explicitly (e.g. `session$user`).
`temp_low_limit`	Lower bound (C) for plausible barometric temperatures. Values outside `[temp_low_limit, temp_hi` are set to `NA` in `airtemp_C_adj` and flagged.
`temp_high_limit`	Upper bound (C) for plausible barometric temperatures.
`pressure_low_kpa`	Lower bound (kPa) for plausible barometric pressure. Values below this are set to `NA` in `airpress_kPa_adj` and flagged. Default is 51 kPa.
`pressure_high_kpa`	Upper bound (kPa) for plausible barometric pressure. Values above this are set to `NA` in `airpress_kPa_adj` and flagged. Default is 112 kPa.
`spike_threshold_kpa`	Threshold (kPa) for flagging sudden pressure spikes between consecutive timesteps (absolute difference in kPa). Default is 1.5 kPa.
`flatline_n`	Integer; number of consecutive identical values required to flag a potential flatline in the pressure series. Default is 6.

Details

If airpress_kPa_adj or airtemp_C_adj do not exist, they are created as copies of the raw airpress_kPa / airtemp_C columns and then modified by the QA/QC rules. Edit flags (OUTLIER_AP, OUTLIER_AT, FLAG_DISTURBANCE, FLAG_FLATLINE) are created if missing and set to TRUE where each rule applies.

For each rule that triggers, a corresponding QA/QC log entry is appended via make_qaqc_log_row() and qaqc_log_append(), using metric "BARO" and field "airpress_kPa_adj".

Value

A data frame containing only rows for select_station, sorted by timestamp, with updated *_adj columns and edit flags. The input is not modified in-place.

See Also

barometric_qaqc_all(), add_nearest_baro()

---

qc_conductivity *Automatic QAQC for conductivity logger data*

Description

Applies a suite of automated quality-control rules to conductivity and water temperature time series for a single station. Raw values are never modified; instead, adjusted copies (conduct_uScm_adj, watertemp_C_adj) are created (or updated if they already exist) and Boolean flag columns are added for review.

Usage

```
qc_conductivity(
  input_data,
  select_station,
  log_root = NULL,
  user = NULL,
  temp_low_limit_C = -2,
  temp_high_limit_C = 40,
  temp_ice_threshold_C = 0.3,
  disturbance_threshold_uScm = 200,
  dry_threshold_uScm = 10,
  air_water_diff_threshold_C = 2
)
```

Arguments

input_data	A data frame of conductivity logger observations. Must contain: site_station_code, timestamp (POSIXct or parseable by lubridate::ymd_hms()), conduct_uScm, and watertemp_C. Optionally airtemp_C (used for the dry inference rule). Typically the output of conductivity_qaqc_all or bind_hobo_files.
select_station	Character scalar. The site_station_code to process. Only rows matching this station are retained.

log_root Character scalar or NULL. Root directory for QAQC log files. When non-NULL, a log CSV is written via qaqc_log_maybe_append(). When NULL (default, Shiny mode), no file is written.

user Character scalar. User name recorded in the QAQC log. Defaults to NULL, which resolves to Sys.info()[["user"]] (the local OS user). In Shiny apps, pass the logged-in user explicitly (e.g. session$user).

temp_low_limit_C Numeric. Lower bound for acceptable water temperature. Values below this are set to NA. Default -2.

temp_high_limit_C Numeric. Upper bound for acceptable water temperature. Values above this are set to NA. Default 40.

temp_ice_threshold_C Water temperature (C) below which ice conditions are flagged. Default 0.3.

disturbance_threshold_uScm Numeric. Absolute conductivity change between consecutive samples that triggers the spike flag. Default 200.

dry_threshold_uScm Numeric. Conductivity below this value is considered a probable dry sensor. Default 10.

air_water_diff_threshold_C Numeric. Maximum absolute air–water temperature difference to support the dry-sensor inference (requires airtemp_C column). Default 2.

Details

The following rules are applied:

FLAG_DISTURBANCE Consecutive conductivity change exceeding disturbance_threshold_uScm; flagged for review (no automatic removal).

FLAG_ICE Water temperature < 0.1 degrees C; flagged for possible ice conditions.

OUTLIER_WT Water temperature between -1 and 0 degrees C corrected to 0.

FLAG_RANGE Water temperature outside temp_low_limit_C to temp_high_limit_C; set to NA.

FLAG_DRY Sliding 12-point window where conductivity <= dry_threshold_uScm and air–water temperature difference <= air_water_diff_threshold_C; flagged as probable dry sensor.

Value

The filtered data frame for select_station, sorted by timestamp, with the following additions:

- **Adjusted columns:** conduct_uScm_adj, watertemp_C_adj.
- **Flag columns:** FLAG_DISTURBANCE, FLAG_ICE, OUTLIER_WT, FLAG_DRY, FLAG_RANGE.
- **Attribute:** attr(result, "qaqc_log") – a tibble of QAQC log rows summarising every rule that fired.

See Also

conductivity_qaqc_all (wrapper that runs this function across all conductivity stations), qc_waterlevel (analogous QAQC for water level), qc_dissox (analogous for dissolved oxygen), wq_drift_correction (downstream drift correction)

qc_dissox *Automatic QAQC for dissolved oxygen logger data*

Description

Applies a suite of automated quality-control rules to dissolved oxygen time series for a single station. Raw values are never modified; instead, adjusted copies (do_mgl_adj, do_percsat_adj, watertemp_C_do_adj) are created (or updated if they already exist) and Boolean flag columns are added for review.

Usage

```
qc_dissox(
  input_data,
  select_station,
  do_high_limit_mgl = 30,
  do_low_limit_mgl = 0,
  temp_ice_threshold_C = 0.3,
  log_root = NULL,
  user = NULL
)
```

Arguments

input_data	A data frame containing dissolved oxygen observations. Must include the columns: site_station_code, timestamp (POSIXct or parseable by lubridate::ymd_hms()), do_mgl, do_percsat, watertemp_C, and airtemp_C. Typically the output of convert_do_mgl_percsat or add_nearest_baro.
select_station	Character scalar. The site_station_code to process. Only rows matching this station are retained. @param temp_ice_threshold_C Water temperature (C) below which ice conditions are flagged. Default 0.3.
log_root	Character scalar or NULL. Root directory for QAQC log files. When non-NULL, a log CSV is written via qaqc_log_maybe_append(). When NULL (default, Shiny mode), no file is written.
user	Character scalar. User name recorded in the QAQC log. Defaults to NULL, which resolves to Sys.info()[["user"]] (the local OS user). In Shiny apps, pass the logged-in user explicitly (e.g. session$user).

Details

The following rules are applied in order:

ERROR_DO Logger error code (-888.88) in DO or water temperature – DO and percent sat set to NA.

OUTLIER_DO DO between 0 and -1 mg/L corrected to 0; DO less than -1 mg/L or greater than 30 mg/L removed (NA).

OUTLIER_WT Water temperature below -1 degrees C removed (NA); between -1 and 0 degrees C corrected to 0.

FLAG_RANGE Sliding 12-point window where DO mg/L is greater than the calibration range of 20 mg/L.

FLAG_ICE Water temperature < 0.3 degrees C flagged for possible ice conditions.

FLAG_DRY Sliding 12-point window where percent sat >= 100 and air–water temperature difference <= 2 degrees C, indicating a potentially dry sensor.

Value

The filtered data frame for select_station, sorted by timestamp, with the following additions:

- **Adjusted columns:** do_mgl_adj, do_percsat_adj, watertemp_C_do_adj.
- **Flag columns:** ERROR_DO, OUTLIER_DO, OUTLIER_WT, FLAG_RANGE, FLAG_ICE, FLAG_DRY.
- **Attribute:** attr(result, "qaqc_log") – a tibble of QAQC log rows summarising every rule that fired.

See Also

qc_waterlevel (analogous QAQC for water level), qc_conductivity (analogous for conductivity), adjust_logger_NA (downstream manual adjustment), convert_do_mgl_percsat (upstream DO conversion)

---

qc_waterlevel *Automatic QAQC for water level logger data*

Description

Applies a suite of automated quality-control rules to water level and water temperature time series for a single station. Raw values are never modified; instead, adjusted copies (waterlevel_m_adj, watertemp_C_adj) are created and Boolean flag columns are added for review.

Usage

```
qc_waterlevel(
  input_data,
  select_station,
  temp_ice_threshold_C = 0.3,
  temp_high_dry_C = 19,
  wl_disturb_threshold_m = 0.1,
  log_root = NULL,
  user = NULL
)
```

Arguments

input_data	A data frame of water level observations. Must contain the columns: site_station_code, timestamp (POSIXct or parseable by lubridate::ymd_hms()), waterlevel_m, watertemp_C, waterpress_kPa, airpress_kPa, and airtemp_C. Typically the output of convert_waterlevel_kPa_m or add_nearest_baro.
select_station	Character scalar. The site_station_code to process. Only rows matching this station are retained.
temp_ice_threshold_C	Water temperature (C) below which ice conditions are flagged. Default 0.3.

`log_root` Character scalar or `NULL`. Root directory for QAQC log files. When non-`NULL`, a log CSV is written via `qaqc_log_maybe_append()`. When `NULL` (default, Shiny mode), no file is written.

`user` Character scalar. User name recorded in the QAQC log. Defaults to `NULL`, which resolves to `Sys.info()[["user"]]` (the local OS user). In Shiny apps, pass the logged-in user explicitly (e.g. `session$user`).

Details

The following rules are applied:

`NEGATIVE_WT` Water temperature below `-1` degrees C removed (`NA`); between `-1` and 0 degrees C corrected to 0.

`FLAG_ICE` Water temperature < 0.3 degrees C flagged for possible ice conditions.

`FLAG_DISTURBANCE` Consecutive water level step change > 0.1 m flagged as potential logger disturbance.

`FLAG_DRY` Possible dry sensor: water temperature > 19 degrees C, water pressure <= air pressure.

Value

The filtered data frame for `select_station`, sorted by `timestamp`, with the following additions:

- **Adjusted columns:** `waterlevel_m_adj`, `watertemp_C_adj`.
- **Flag columns:** `FLAG_DISTURBANCE`, `OUTLIER_WT`, `FLAG_ICE`, `FLAG_DRY`.
- **Attribute:** `attr(result, "qaqc_log")` – a tibble of QAQC log rows summarising every rule that fired.

See Also

`qc_dissox` (analogous QAQC for dissolved oxygen), `qc_conductivity` (analogous for conductivity), `adjust_waterlevel_spike` (downstream spike removal), `adjust_logger_NA` (downstream manual adjustment), `convert_waterlevel_kPa_m` (upstream WL conversion)

`qc_watertemp` *QA/QC a single water temperature time series*

Description

Applies automatic QA/QC rules to a water temperature time series for one station: flags possible ice and dry conditions, corrects small negative temperatures to 0, and removes clearly impossible values (< -1 C). Returns the input data (filtered to `select_station`) with adjusted columns and boolean flag/edit columns.

Usage

```
qc_watertemp(
  input_data,
  select_station,
  temp_ice_threshold_C = 0.3,
  temp_high_dry_C = 19,
  temp_air_water_diff_C = 2,
  log_root = NULL,
  user = NULL
)
```

Arguments

input_data	Data frame of water-level or dissolved-oxygen logger data containing at least timestamp, site_station_code, and watertemp_C. An airtemp_C column is optional: when present, air-water comparison dry checks are also applied.
select_station	Character, station code to process (matches site_station_code).
temp_ice_threshold_C	Water temperature (C) below which ice conditions are flagged. Default 0.3.
temp_high_dry_C	Water temperature (C) above which a dry sensor is suspected. Default 19.
temp_air_water_diff_C	Absolute difference (C) between water and air temperature below which a dry sensor is suspected (when airtemp_C is present). Default 2.
log_root	Character scalar or NULL. Root directory for QAQC log files. When non-NULL, a log CSV is written via qaqc_log_maybe_append(). When NULL (default, Shiny mode), no file is written.
user	Character username written into the QA/QC log. Defaults to NULL, which resolves to Sys.info()[["user"]].

Details

The following rules are applied:

OUTLIER_WT Water temperature < -1 C: set to NA in watertemp_C_adj.

OUTLIER_WT Water temperature in [-1, 0) C: corrected to 0 in watertemp_C_adj.

FLAG_ICE Water temperature < temp_ice_threshold_C: flagged for review; data not changed.

FLAG_DRY Possible dry sensor: water temperature > temp_high_dry_C; or, when airtemp_C is present, air-water temperature within temp_air_water_diff_C degrees C, or air warmer than water.

Value

A data frame containing only rows for select_station, sorted by timestamp, with watertemp_C_adj, FLAG_ICE, FLAG_DRY, OUTLIER_WT, and OUTLIER_WT columns added. attr(result, "qaqc_log") holds a tibble of QA/QC log rows.

`resample_timestamps` *Align (snap) logger timestamps to a regular interval*

Description

This function standardises the `timestamp` column onto a regular time grid (e.g., 5, 10, 15, 30 minutes, 1 hour) using rounding, flooring, or ceiling. The original logger times are preserved in `timestamp_raw` (created once if missing).

Usage

```
resample_timestamps(
  input_data,
  interval = "15 min",
  align = c("round", "floor", "ceiling"),
  select_station = NULL,
  metric = NULL,
  log_root = NULL,
  user = NULL
)
```

Arguments

`input_data`	A data frame containing at least a `timestamp` column. If `select_station` is used, a `site_station_code` column is also required.
`interval`	Character string giving the target time interval, passed to `lubridate::round_date()`, e.g. `"5 min"`, `"10 min"`, `"15 min"`, `"30 min"`, `"1 hour"`.
`align`	How to align timestamps to the grid. One of `"round"`, `"floor"`, `"ceiling"`. See `lubridate::round_date()`.
`select_station`	Optional station code. If supplied and `site_station_code` exists, the function will operate only on that station's rows. The full input data frame (all stations) is always returned.
`metric`	Optional character code for the metric (e.g. `"WL"`, `"DO"`, `"COND"`, `"BARO"`), used only for QA/QC logging.
`log_root`	Optional path to the root QA/QC log folder. If provided together with `metric` and `select_station`, a log entry is appended for this operation.
`user`	Username recorded in QA/QC logs. Defaults to `NULL`, which resolves to `Sys.info()[["user"]]` (the local OS user). In Shiny apps, pass the logged-in user explicitly (e.g. `session$user`).

Details

Importantly, this function **does not change the sampling frequency**: it does not drop or average rows. If your logger is at 5-minute resolution and you specify `interval = "30 min"`, all 5-minute rows are kept; their timestamps are simply moved to the nearest 30-minute grid line. This may result in multiple rows sharing the same `timestamp`.

For true resampling/aggregation (e.g., converting 5-minute data to 30-minute means), use a separate aggregation function.

If `log_root`, `metric`, and `select_station` are provided, a QA/QC log entry is written using the package logging helpers.

Value

A data frame with:

- `timestamp_raw`: original logger timestamps (created if missing).
- `timestamp`: snapped timestamps on the specified grid.

All other columns are returned unchanged.

Examples

```
## Not run:
aligned_wl <- resample_timestamps(
  input_data     = wl_bound,
  interval       = "15 min",
  align          = "round",
  select_station = "ALBR_ST_30",
  metric         = "WL",
  log_root       = "data/processed"
)

## End(Not run)
```

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wl_multipoint_correction

Apply multi-point linear drift correction to water level

Description

This function applies a piecewise-linear drift correction to `waterlevel_m_adj` for a single station. You provide a set of time-stamped drift control points , each with a desired drift value in metres. The function linearly interpolates between successive control points and adds that correction to `waterlevel_m_adj`. Data before the first control point and after the last control point are left unchanged.

Usage

```
wl_multipoint_correction(
  input_data,
  select_station,
  log_root = NULL,
  drift_points,
  user = NULL,
  manual_note = NULL
)
```

Arguments

`input_data`	Data frame containing at least: `site_station_code`, `timestamp`, `waterlevel_m` or `waterlevel_m_adj`.
`select_station`	Character. Station code to correct.

`log_root` Character. Root folder for QA/QC logs.

`drift_points` A data frame or tibble with columns:

timestamp Drift control timestamps (POSIXct or character). Character strings are parsed in the logger's timezone (detected from the POSIXct `tzone` attribute of `input_data$timestamp` or an optional `timezone` column). Falls back to `"UTC"` if neither source is available. Pass a POSIXct with the correct `tzone` to override this detection.

drift_m Numeric drift (m) at each timestamp. Positive means the logger is reading too low (we will add this amount to bring the series up).

Must contain at least two rows.

`user` Character. Username for the QA/QC log. Defaults to `NULL`, which resolves to `Sys.info()[["user"]]` (the local OS user). In Shiny apps, pass the logged-in user explicitly (e.g. `session$user`).

`manual_note` Optional character note to include in the log.

Details

For each interval $[t_i, t_{i+1}]$ between control points, a linear correction is constructed:

$$c(t) = d_i + (d_{i+1} - d_i) * \frac{t - t_i}{t_{i+1} - t_i}$$

where d_i and d_{i+1} are the drift values (in metres) at times t_i and t_{i+1}. The adjusted water level is then:

$$waterlevel_m_adj = waterlevel_m_adj + c(t)$$

No correction is applied outside the range of the earliest and latest control timestamps.

A single QA/QC log row is written with code `MULTI_WL_DRIFT` summarising the time window and drift values.

Value

A modified data frame with updated `waterlevel_m_adj` and a new logical column `edit_wl_drift` (`TRUE` for every row where a non-zero correction was applied) for `select_station`. All other stations/rows are unchanged.

`wl_set_datum` *Set an arbitrary datum for water-level time series*

Description

Creates a consistent arbitrary elevation system for a station by anchoring the first valid levelling visit's primary benchmark (PBM) at a user-specified elevation (default 100.0 m) and translating all other benchmarks using within-visit survey differences. If later visits use a different PBM, its arbitrary elevation is inferred from the existing benchmark network (it is *not* forced to 100.0 m).

Usage

```
wl_set_datum(
  input_data,
  level_runs_path = NULL,
  select_station,
  log_root = NULL,
  user = NULL,
  level_runs = NULL,
  max_anchor_gap_h = 1,
  bm_tolerance_m = 0.02,
  datum_elev_m = 100
)
```

Arguments

input_data	A data frame of logger observations (one or more stations). Must contain site_station_code, timestamp (POSIXct or parseable), and either waterlevel_m_adj (preferred) or waterlevel_m.
level_runs_path	Character scalar or NULL. Path to a levelling-runs CSV file. Required when level_runs is NULL.
select_station	Character scalar. The site_station_code to process.
log_root	Character scalar or NULL. Root directory for QAQC log files. When non-NULL, a log entry is written via qaqc_log_maybe_append. When NULL (default, Shiny mode), no file is written.
user	Character scalar. Analyst name recorded in the QAQC log. Defaults to NULL, which resolves to Sys.info()[["user"]] (the local OS user). In Shiny apps, pass the logged-in user explicitly (e.g. session$user).
level_runs	A data frame of levelling-run data (alternative to level_runs_path for Shiny mode). Exactly one of level_runs_path or level_runs must be supplied.
max_anchor_gap_h	Numeric scalar. Maximum allowable gap (in hours) between the anchor visit datetime and the nearest logger sample. Default 1. Increase for loggers with coarse time steps (e.g. 2-hour intervals).
bm_tolerance_m	Numeric scalar. Benchmark inconsistency threshold (in metres). A warning is issued if the inferred absolute elevation for a benchmark differs by more than this amount across visits. Default 0.02.
datum_elev_m	Numeric scalar. The arbitrary elevation (in metres) assigned to the first valid visit's primary benchmark. Default 100. Adjust if you want a different datum origin (e.g. a known geodetic elevation).

Details

A single constant datum offset is computed from the first valid visit's mean water-surface elevation and applied to the station's logger stage series, producing the column wl_elev_m.

Level-runs data format:

The level-runs CSV (or data frame) must contain at minimum:

- site_station_code
- visit_date (e.g. "2025-04-19" or "4/19/2025")

- visit_time (e.g. "16:26" or "16:26:00")
- primary_bm – must exactly match a BM* or TBM* column name (e.g. "BM1", "TBM1")
- grade – one of "Excellent", "Good", "OK", "Poor"
- One or more benchmark columns named BM* or TBM*
- One or more water-surface columns named water_elev_<n> (e.g. water_elev_1, water_elev_2)

All survey values are assumed to be in the same per-visit coordinate system. The absolute arbitrary elevation for any value is: pbm_abs + (value_raw - pbm_raw).

Grade handling: No filtering by grade is performed, but a warning is issued if the first valid visit (the datum anchor) is graded "Poor".

Benchmark network: The function iterates visits in chronological order. If a visit's PBM is not yet in the network, it is inferred from any co-observed benchmark that *is* known. An error is raised if a visit cannot be connected to the existing network. Benchmark inconsistencies exceeding bm_tolerance_m across visits trigger a warning.

Timezone handling: Visit date/time strings are parsed in the logger's timezone (detected from the POSIXct tzone attribute of input_data$timestamp or an optional timezone column). Falls back to "UTC" if neither is available. Pass logger timestamps as POSIXct with the correct tzone to ensure accurate matching.

Value

The full input_data data frame with two columns added/updated for select_station:

- wl_elev_m – water level on the arbitrary datum (m).
- flag_wl_datum_applied – logical flag (TRUE for station rows where the datum was applied).

attr(result, "qaqc_log") contains a tibble with the WL_DATUM_APPLIED log entry.

See Also

qc_waterlevel (upstream automatic QAQC), convert_waterlevel_kPa_m (upstream kPa-to-m conversion), wl_multipoint_correction (water level drift correction)

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wq_drift_correction *Water-quality drift correction (COND / DO / pH)*

Description

Apply forward linear drift correction to a water-quality time series using discrete reference measurements collected during field visits. The correction ramps linearly from zero at the start of each segment to the full correction at the visit endpoint, then restarts at the next segment. No correction is applied after the last visit.

Usage

```
wq_drift_correction(
  input_data,
  ref_data_path = NULL,
  select_station,
  metric,
  method = c("ratio", "offset"),
  max_ref_gap_h = 1,
  log_root = NULL,
  user = NULL,
  ref_data = NULL
)
```

Arguments

input_data	Data frame of QA/QC'd logger data. Must contain site_station_code, timestamp (POSIXct or parseable), and the adjusted value column for the chosen metric (conduct_uScm_adj, do_percsat_adj, or ph_adj).
ref_data_path	Character path to a CSV file containing reference measurements. Set to NULL when providing ref_data directly (e.g., in a Shiny app). One of ref_data_path or ref_data must be provided.
select_station	Character. The site_station_code to correct. Only rows matching this station are modified; all other rows are returned unchanged.
metric	Character identifying the water-quality metric. Accepted values (case-insensitive): "conductivity", "cond", "dissolvedoxygen", "do", "ph".
method	One of "ratio" (multiplicative, default) or "offset" (additive).
max_ref_gap_h	Numeric. Maximum allowable time gap (in hours) between a reference measurement timestamp and the nearest logger sample. Reference events exceeding this gap are skipped with a warning. Default 1.
log_root	Optional path to the QA/QC log folder. If provided, a log entry is written to disk. If NULL, log rows are still returned as an attribute on the result.
user	Character. Username recorded in QA/QC logs. Defaults to NULL, which resolves to Sys.info()[["user"]] (the local OS user). In Shiny apps, pass the logged-in user explicitly (e.g. session$user).
ref_data	Data frame of reference measurements, as an alternative to ref_data_path. Must contain the columns described in Details.

Details

Two correction methods are available:

- **ratio** (multiplicative): each value is scaled by a linearly interpolated ratio between 1.0 (segment start) and the reference/logger ratio (segment end).
- **offset** (additive): each value is shifted by a linearly interpolated offset between 0 (segment start) and the reference minus logger difference (segment end).

Two event types control how segment boundaries are determined:

- **check**: the end-target is the nearest logger sample to the visit timestamp; the next segment restarts at that same sample.

- **calibration**: the end-target is the last logger sample strictly before `calibration_datetime`; the next segment restarts at the first sample at or after `calibration_datetime` (post-calibration values are trusted).

Reference data format

The reference data (CSV or data frame) must contain these columns:

metric Character. The water-quality metric (e.g., `"conductivity"`, `"do"`, `"ph"`).

site_station_code Character. Station identifier matching the logger data.

timestamp Character or POSIXct. When the reference measurement was taken.

watertemp_C Numeric. Water temperature at the time of measurement.

value Numeric. The reference measurement value.

unit Character. Unit of `value` (e.g., `"uS/cm"`, `"%sat"`, `"pH"`).

event_type Character. One of `"check"`, `"calibration"`, or `"ignore"`.

calibration_datetime Character or POSIXct. Required for `"calibration"` events; the moment the sensor was recalibrated. `NA` or empty for other event types.

Timezone handling

Reference timestamps are parsed into the logger's timezone, which is determined from the POSIXct `tzone` attribute or an optional `timezone` column in `input_data`. Falls back to `"UTC"` if neither is available.

Value

The input data frame with drift-corrected values written back to the adjusted column (`*_adj`), plus an `edit_*_drift` logical flag column (`TRUE` for corrected rows). For dissolved oxygen, `do_mgl_adj` is also corrected using the same ratio as `do_percsat_adj` (ratio is dimensionless and applies consistently across both units). A `"qaqc_log"` attribute is attached containing the log rows as a tibble (or `NULL` if no rows were corrected).

See Also

`wl_multipoint_correction` (water level drift correction)

`write_clean_data` *Write final clean data files (v1.0)*

Description

Subsets QAQC'd logger data to the columns relevant for a given metric, splits by station and year, and writes per-station per-year v1.0 CSV files to a `<year>/clean/` directory. This is the final step in the processing pipeline, typically run after all QAQC adjustments (e.g. `qc_waterlevel`, `qc_dissox`) are complete.

Usage

```
write_clean_data(input_data, path_to_output_folder = NULL, metric)
```

Arguments

input_data	A data frame of QAQC'd logger observations. Must contain at minimum site_station_code, timestamp, and the metric-specific required columns listed under **Details**. Any columns matching flag_* or edit_* are automatically retained alongside the metric-specific columns.
path_to_output_folder	Character scalar or NULL. Root directory where per-year v1.0 CSVs will be written. When NULL (default, Shiny mode), no files are written; the function still validates columns and returns input_data invisibly.
metric	Character scalar identifying the data type. Must be one of "dissolvedoxygen", "barometric", "waterlevel", "watertemp", "airtemp", or "conductivity". Controls which columns are required and the file-name prefix (see **Details**).

Details

Each metric maps to a set of required and optional columns, plus a file-name prefix:

"dissolvedoxygen" **(DO)** Required: do_mgl, do_mgl_adj, do_percsat, do_percsat_adj. Optional: watertemp_C_do, watertemp_C_do_adj.

"barometric" **(BARO)** Required: airpress_kPa, airpress_kPa_adj. Optional: airtemp_C, airtemp_C_adj.

"waterlevel" **(WL)** Required: waterlevel_m, waterlevel_m_adj. Optional: watertemp_C, watertemp_C_adj, waterlevel_reference.

"watertemp" **(WT)** Required: watertemp_C, watertemp_C_adj. No optional columns.

"airtemp" **(AT)** Required: airtemp_C. Optional: airtemp_C_adj.

"conductivity" **(CO)** Required: conduct_uScm. Optional: conduct_uScm_adj, spc_uScm_adj, watertemp_C, watertemp_C_adj.

All metrics also require site_station_code and timestamp. Output files are written to <path_to_output_folder>/<y

If input_data contains a metric column whose value differs from the metric argument, a warning is issued.

Value

input_data returned invisibly (unmodified), allowing the function to be used at the end of a pipe chain.

See Also

get_logger_data (reads the v1.0 files produced by this function), qc_waterlevel, qc_dissox, qc_barometric (upstream QAQC functions whose output is typically passed here)